Q) What is a dangling pointer? TN

Ans) A dangling pointer is a pointer that is pointing to a memory location that has been deleted or deallocated. Dangling pointers arise during object destruction, when an object that has an incoming reference is deleted or freed, without modifying the value of the pointer, so that the pointer still points to the memory location of the deallocated memory.

Example:-

```
void main()
{
        int *ptr = (int *) malloc (sizeof(int));
        free (ptr); // After this call, ptr becomes
                                a dangling pointer
}
```

Q) What is a void pointer?

Ans) void pointer is a specific pointer type that points to some data location in storage, which does not have any specific type. The type of data that it points to can be any. If we assign address of char data type to void pointer, then it will become char pointer. If we assign address of int data type to void pointer, then it will become int pointer and so on.

Example:-

```
void main()
{
        int a=10;
        char c='x';
        void *ptr; //void pointer
        ptr=&a; //ptr changes to integer pointer as address of integer
                                is assigned to it
        ptr=&c; //ptr now changes to character pointer
}
```

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Memory

Q) What is a Null pointer?

Ans) A null pointer is a pointer which is not pointing to anything i.e. it is assigned a null value. ~~In case w~~ When we do not have address to be assigned to a pointer, it is a good practice to assign it to null. So, null pointer is a special reserved value of a pointer. Every pointer type like int *, char *, etc. have such a reserved value.

Example

```
void main()
{
    int *ptr= NULL;
    printf("\n The value of ptr is %u", ptr);
    getch();
}
```

TN

Output

The value of ptr is 0.

Thus, null pointer is a value assigned to a pointer of any data type whereas a void pointer is a data type which remains void as long as an address of a data type is not assigned to it.

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